

SOSUNOVO, Russian Federation

WMO: 318660

Lat: **46.5456N**Lon: **138.3381E**Elev: **14**StdP: **14.69**Time Zone: **10.00 (E10)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.3	-5.2	-24.5	1.4	-3.3	-21.8	1.7	-1.0	37.7	21.2	30.9	15.6	7.7	260	0.692

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.1	76.6	66.4	72.4	64.5	68.9	62.8	68.0	73.5	66.1	70.2	64.5	67.8	8.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	96.7	70.0	64.5	91.0	68.2	63.0	86.0	66.3	32.5	73.5	31.0	70.4	29.7	68.1	73.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.2	22.4	19.2	-16.8	85.2	3.3	3.3	-19.2	87.6	-21.1	89.5	-23.0	91.4	-25.4	93.8		
(5)			-17.3	72.4	3.1	0.9	-19.5	73.1	-21.3	73.6	-23.1	74.1	-25.4	74.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.1	10.5	14.2	24.6	35.6	44.0	50.9	59.8	63.3	56.4	43.2	27.3	13.8
	DBStd	19.14	7.64	7.99	6.88	5.21	6.14	4.88	4.87	3.94	5.23	5.90	9.66	7.97
	HDD50	5735	1224	1001	787	435	211	41	0	0	10	221	681	1124
	HDD65	10234	1689	1421	1252	883	652	422	177	80	262	676	1131	1589
	CDD50	1024	0	0	0	2	24	70	303	414	201	10	0	0
	CDD65	47	0	0	0	0	0	1	15	28	3	0	0	0
	CDH74	243	0	0	0	0	5	5	117	105	11	0	0	0
	CDH80	40	0	0	0	0	1	0	30	9	0	0	0	0
Wind	WSAvg	9.2	10.7	9.9	9.4	8.8	9.2	8.3	7.3	7.5	8.6	9.1	9.9	11.1
Precipitation	PrecAvg	29.5	0.6	0.4	1.1	1.7	3.5	2.8	4.3	4.9	4.2	3.0	1.7	1.1
	PrecMax	54.8	2.4	1.3	3.9	9.6	6.1	18.3	12.8	20.6	11.7	9.5	9.1	3.5
	PrecMin	9.9	0.0	0.0	0.2	0.2	0.2	0.3	0.5	0.6	0.6	0.2	0.0	0.0
	PrecStd	12.9	0.6	0.3	0.9	1.8	1.5	3.3	3.1	4.9	3.3	2.2	2.4	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.0	38.6	49.0	62.8	70.6	72.5	84.4	80.9	75.7	65.4	53.6	37.8
		MCWB	30.0	29.7	38.2	47.8	53.6	63.6	69.3	68.0	64.3	52.1	42.0	34.3
	2%	DB	28.5	32.8	42.2	53.7	64.2	66.0	76.6	77.0	71.1	60.0	47.4	32.3
		MCWB	25.1	27.4	33.3	42.5	50.5	58.7	66.2	67.2	61.2	50.4	39.4	28.6
	5%	DB	25.3	29.9	37.4	48.6	58.8	61.7	70.4	73.2	67.7	56.7	43.8	29.1
		MCWB	21.6	26.1	31.0	39.2	48.6	57.2	63.5	66.4	59.9	48.9	39.0	25.6
	10%	DB	22.4	27.2	34.3	44.6	53.3	57.9	65.6	69.9	65.4	54.2	41.1	26.3
		MCWB	19.2	23.4	29.8	36.8	46.6	54.0	62.5	64.8	59.4	48.1	37.1	23.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.8	31.2	38.9	47.6	55.3	64.3	70.8	70.7	66.1	55.8	44.3	34.1
		MCDB	31.3	34.9	48.6	60.5	66.1	71.5	81.4	75.8	72.1	59.6	47.6	36.1
	2%	WB	25.1	28.7	34.3	43.3	51.8	60.6	66.9	68.9	63.8	53.1	41.9	29.4
		MCDB	27.9	32.1	39.9	52.4	61.4	64.4	74.8	73.6	68.0	57.2	44.9	31.2
	5%	WB	22.1	26.3	32.1	40.2	49.1	57.1	64.4	67.2	62.3	51.4	40.0	26.0
		MCDB	24.8	29.8	35.6	46.6	57.8	60.8	68.0	71.1	65.9	54.8	43.4	28.5
	10%	WB	19.5	23.6	30.6	37.9	46.8	54.3	62.9	65.8	60.5	49.3	37.1	23.0
		MCDB	22.6	27.3	33.9	43.1	52.2	57.3	65.6	69.0	64.0	53.4	40.8	26.1
Mean Daily Temperature Range	5% DB	MDBR	15.8	17.7	15.3	12.8	11.9	8.9	8.3	10.1	15.6	18.0	15.8	15.7
		MCDBR	17.0	19.0	18.2	21.8	23.2	18.1	20.3	17.8	19.0	23.0	18.8	17.0
		MCWBR	14.6	15.1	11.8	12.6	11.9	11.6	11.2	9.5	11.1	14.8	14.5	14.5
	5% WB	MCDBR	15.8	17.2	15.3	19.8	20.9	17.2	14.9	13.5	14.2	16.4	16.5	16.0
		MCWBR	14.2	14.0	10.7	12.1	11.8	11.8	9.3	8.5	9.8	12.9	14.0	14.6
Clear-Sky Solar Irradiance	taub	0.306	0.356	0.428	0.515	0.546	0.524	0.484	0.452	0.406	0.404	0.358	0.320	
	taud	2.204	2.052	1.915	1.807	1.828	1.988	2.169	2.270	2.313	2.148	2.174	2.159	
	Ebn at Noon	242	248	246	235	234	240	249	253	254	229	217	218	
	Edh at Noon	27	39	52	63	64	55	45	40	35	35	28	25	
All-Sky Solar Radiation	RadAvg	525	851	1239	1459	1514	1531	1430	1408	1274	891	575	431	
	RadStd	45	56	112	169	154	219	150	125	117	74	51	35	